2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.201 Databases

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

The following represent tables and sample data in a simple relational database developed for tracking stock levels and loyalty card use across a number of retail outlets.

Customer

CardNum	CustName	CustAddr
1	A. Scott	InfoLab21

StoreStock

StoreNum	StockNum	QtyAvail	Price
1	2	100	53

PurchaseRecord

CardNum	Date	StockNum	Qty	StoreNum
1	6 Jan 16	2	5	1

Store

StoreNum	StoreAddr
1	Lancaster

1.a Define and give examples from the above database for the following terms.

Relation, Tuple, Attribute, Domain, Super Key, Relational Database Schema

[**7 marks**]

1.b Produce a *Query Tree* showing the sequence of relational algebra expressions required to produce a list in the following form of stores holding 200 or more items of any product:

Store	StockNum	QtyAvail
Lancaster	2	200
Lancaster	3	400

[**6 marks**]

1.c Produce a set of SQL queries to accomplish the following:

i. A new customer, J.Brookes from Engineering, has been allocated CardNum 2; add this information into the database. (2 marks)

ii. J.Brookes moves to Physics; change the information in the database to reflect this. (2 marks)

iii. Use a sub-Select/ nested Select statements and a Join to find the names of customers that have purchased items from the Lancaster store. (4 marks)

iv. Produce a list of customers and the number of different types of items they purchased on each day using the following output format. (4 marks)

Customer	Purchases	Date

[**12 marks**]

Total 25 marks

Question 2

2.a Explain the following terms used in Relational Databases and give an illustrative example of each.

i. Functional dependency

ii. Full functional dependency

iii. Partial dependency

iv. Transitive dependency **[9 marks]**

2.b Using the terms from question 2a

i. Define *Second Normal Form* and *Third Normal Form*

ii. Identify which of these forms of dependency are not considered when converting from
First Normal Form to *Second Normal Form* **[5 marks]**

2.c The following relation is in *First Normal Form*.

i. Identify why the relation is in *First Normal Form*

ii. With reference to the different types of functional dependencies that exist within the relation and clearly showing each and every step required, convert it to *Third Normal Form.*

You must identify the type of any dependencies you identify and show the state of any tables after completing each step in the process.

Championship	Event	Referee	RefereeName	Participant	ParticipantName	Points
London	Javelin	1	Hughes	100	Smith	24
London	Javelin	1	Hughes	103	Brown	13
London	Discus	1	Hughes	264	Drake	19
London	Shot Put	2	Schmid	302	Suri	12
Paris	Javelin	1	Hughes	100	Smith	20
Paris	Discus	3	Gleyzon	264	Drake	18
Barcelona	Javelin	1	Hughes	100	Smith	22
Barcelona	Discus	1	Hughes	264	Drake	15

Attributes forming the primary key are underlined and you only need to consider dependencies as shown. **[11 marks]**

Total 25 marks

Question 3

3.a A PARTS file with Part# as hash key includes records with the Part# values given in the table below. The file uses 4 buckets, numbered 0 to 3. Each bucket is one disk block and holds two records. Load these records into the file in the given order using the hash function h(K) = K mod 4. To assist you, the results of applying this function are also given in the table below. You should use collision resolution through chaining. You need only show the final state of the hash file.

Part#	h(Part#)
450	2
505	1
206	2
333	1
72	0
249	1
391	3
26	2

[4 marks]

3.b Describe the drawbacks of conventional hashing.

[4 marks]

3.c You have an extendible hash organisation, where each disk block holds 2 records. Load the following 7 records into the hash file.

450, 505, 206, 333, 72, 151, 391

The hash function is to take the lower 9 bits of the binary value of the search field. The results of this function are shown in the table below.

Show each step of the process, and the state of the hash file after each step.

[17 marks]

450	111000010
505	111111001
206	011001110
333	101001101
72	001001000
151	010010111
391	110000111

Total 25 marks

Question 4

4.a

ACTOR

A_ID	A_Name	Age	Film	Signed Contract
MOO	R. Moore	55	Live and Let Die	1973
CON	S. Connery	55	Dr. No	1962
MOO	R. Moore	40	The Man who Haunted Himself	1970
AND	U. Andress	40	Dr. No	1962
CON	S. Connery	62	The Rock	1996
WEA	S. Weaver	45	Aliens	1986
HAR	E. Harris	45	The Abyss	1989

FILM

F_ID	Film_Name	Director
LLD	Live and Let Die	G. Hamilton
DNO	Dr. No	T. Young
MHH	The Man Who Haunted Himself	B. Dearden
ROC	The Rock	M. Bay
ALI	Aliens	J. Cameron
ABY	The Abyss	J. Cameron

DIRECTOR

D_ID	Dir_Name	Location
HAM	G. Hamilton	London
YOU	T. Young	Glasgow
DEA	B. Dearden	Edinburgh
BAY	M. Bay	Chicago
CAN	J. Cameron	Los Angeles

Question 4 continues on the next page...

Question 4 continued…

From the information shown in the tables on the previous page,

i. identify the entity types, the relationships between them, and their attributes. List the instances of the entity types. Give the cardinality of any relationships.

[5 marks]

ii. Convert this information into an ER model diagram. The diagram should clearly indicate primary keys and relationship cardinality.

[4 marks]

iii. Convert the ER diagram into a set of relations.

[8 marks]

Your answers should include the rationale behind your decisions and classifications.

4.b Here are two relations with some sample data.

course

c#	course_name	l#
C1	mathematics	L1
C2	physics	L2

lecturer

l#	lecturer_name
L1	Smith
L2	Jones

Convert the following SQL query into a canonical query tree involving the relational algebra operators.

```
SELECT course_name
FROM course C, lecturer L
WHERE (C.L# = L.L#) and (L.LECTURER_NAME = "Smith");
```

Apply simple heuristics to optimise the tree. Describe the heuristics applied and show the tree after each transformation.

[8 marks]

Total 25 marks

--- END OF PAPER ---